

CLOUD TECH SMART SOLUTIONS

Smart Home System Architecture

System Documentation & Schematics

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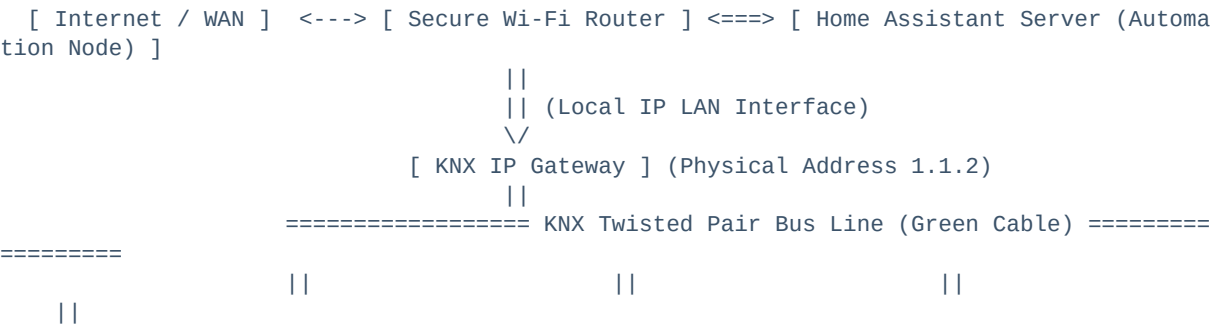
SECTION 3: SMART HOME TOPOLOGY & ACTUATORS

3.1 Core System Architecture

The smart building is engineered on the decentralized **KNX Bus System** to ensure 100% hardware reliability. High-level automations, remote control interfaces, and scheduling profiles are governed by a secure local server running **Home Assistant**.

Addr	Device Type	Channels	System Description	Area / Load
1.1.1	KNX Power Supply 640mA	N/A	KNX Bus Voltage & Diode Power	Main Cabinet
1.1.2	KNX-IP Interface	N/A	Local Network Bridge to Home Assistant	Network Rack
1.1.10	Switch Actuator	12-Chan nel	Lighting Relays & Switched Sockets	Ground Floor Lights
1.1.11	Switch Actuator	12-Chan nel	Lighting Relays & Exhaust Fans	First Floor Lights
1.1.12	Dimming Actuator	4-Chann el	Dimmable LED Spots (Trailing Edge)	Living / Dining Rooms
1.1.13	Shutter Actuator	8-Chann el	Roller Blind Motors & Curtains	Windows / Balconies
1.1.14	Heating Actuator	6-Chann el	Underfloor Valve Actuator (24V electrothermal)	Climate Loops
1.1.20	Motion & Lux Sensor	4-Chann el	Ceiling Presence-Lux Sensor integration	Corridors & Bathrooms
1.1.30	KNX Push Button Taster	6-Button	Wall glass switch switches with temp sensors	Living Rooms

3.2 Topological Data Stream



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[KNX Power Supply] all Glass Switches] (1.1.30)	(1.1.1) (1.1.10 - 1.1.11)	[Relay Actuators] (1.1.10 - 1.1.11)	[Shutter Controller] [W (1.1.13)